Diabetes Classification and Clustering Analysis

1. Introduction

The objective of this project was to build both supervised and unsupervised machine learning models to analyze and predict diabetic conditions from a structured clinical dataset containing 1000 patient records.

The classification task involved predicting one of three classes:

- N = Normal
- P = Predict-diabetic
- Y = Diabetic

In addition to supervised classification, unsupervised clustering was performed to analyze the natural grouping of patients without using labels.

- The final solution includes:
- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Classification using Logistic Regression and Random Forest
- Hyperparameter tuning
- Cross-validation
- Clustering using KMeans
- PCA visualization
- Model serialization for deployment readiness

The Random Forest model achieved strong performance with high accuracy across folds. Clustering analysis revealed that the dataset naturally forms two dominant groups, suggesting that predict-diabetic cases lie in a transitional region between normal and diabetic patients.

2. Problem Statement

This assignment involves analyzing a multi-class diabetes dataset containing three categories: N (Non-diabetic), P (Predict-diabetic), and Y (Diabetic). The dataset consists of clinical and biochemical features such as HbA1c, BMI, cholesterol levels, triglycerides, urea, creatinine and other medically relevant indicators that are commonly associated with diabetes diagnosis and progression.

The task is to apply machine learning techniques to both predict diabetic class labels and examine the underlying structure of the data through clustering. The specific goals of this assessment are:

1. To develop a classification model capable of accurately predicting the diabetic class (N, P, Y) for a given patient.
2. To implement a clustering algorithm to identify natural groupings in the dataset without using target labels.
3. To compare supervised classification results with unsupervised clustering behavior.
4. To provide analytical interpretation, error analysis, and insights derived from the models.

3. Dataset Overview

The dataset consists of structured clinical records collected from 1000 patients. Each record contains demographic information and multiple biochemical measurements commonly used in diabetes diagnosis and monitoring.

3.1 Dataset Size and Target Variable

The dataset contains structured patient records and a clearly defined target variable used for supervised learning. The key structural details are summarized below.

- Total Records: 1000
- Target Variable: CLASS
- Prediction Type: Multi-class classification

The target variable includes three categories:

- N = Normal
- P = Predict-diabetic
- Y = Diabetic

<u>Class Distribution:</u>

After removing whitespace inconsistencies in the CLASS column, the class distribution is as follows:

- Diabetic (Y): 844 samples
- Non-diabetic (N): 103 samples
- Predict-diabetic (P): 53 samples

This indicates a significant class imbalance, with the diabetic class dominating the dataset (~84%). The minority classes (N and P) together represent less than 20% of the data.

Due to this imbalance:

- Accuracy alone is not a reliable evaluation metric.
- Macro and weighted F1-scores are more appropriate.
- Class weighting was applied during model training.
- Stratified splitting was used to preserve distribution across folds.

3.2 Feature Description

The dataset consists of demographic and biochemical measurements that are clinically relevant to diabetes diagnosis and progression.

It includes Gender (categorical: M/F) and AGE (20–79 years), where age is a known risk factor for Type 2 diabetes. Kidney function indicators such as Urea and Creatinine (Cr) are measured in mg/dL and may reflect renal complications often associated with diabetes.

The most important glycemic marker in the dataset is HbA1c (%), which represents long-term blood glucose levels and is widely used as a diagnostic criterion for diabetes classification.

The lipid profile features include Total Cholesterol (Chol), Triglycerides (TG), HDL, LDL and VLDL, all measured in mg/dL. These parameters are linked to metabolic health, insulin resistance, and cardiovascular risk.

Additionally, BMI (Body Mass Index) is included as an indicator strongly associated with obesity and diabetes risk.

The target variable, CLASS, is categorical with three labels: N (Non-diabetic), P (Prediabetic), and Y (Diabetic).

4. Data Cleaning and Preparation

During initial data inspection, minor inconsistencies were identified in the target variable. Specifically, some class labels contained trailing whitespace characters, which could lead to incorrect class grouping during analysis. These inconsistencies were corrected to ensure uniform labeling across all records.

Additionally, identifier-based columns such as ID and No_Pation were removed from the dataset. These columns serve only as unique record identifiers and do not carry predictive information relevant to the modeling task. Retaining such features could introduce unnecessary noise or unintended bias into the model.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the distribution of the target variable, examine feature behavior, detect outliers, and analyze relationships among variables. This step helps identify potential modeling challenges and guides preprocessing decisions.

5.1 Target Variable Distribution

The target variable distribution reveals a significant class imbalance in the dataset. Approximately 84.4% of the records belong to the Diabetic (Y) class, while 10.3% are Non-diabetic (N) and only 5.3% are Predict-diabetic (P). This confirms that the dataset is heavily skewed toward diabetic cases.

To ensure fair model evaluation, a stratified train-test split was applied. The class proportions in the training and testing sets closely match the overall distribution, preserving the imbalance structure. Given this skewed distribution, metrics such as F1-score were considered more appropriate than accuracy for evaluating model performance in this project.

5.2 Feature Distribution Analysis

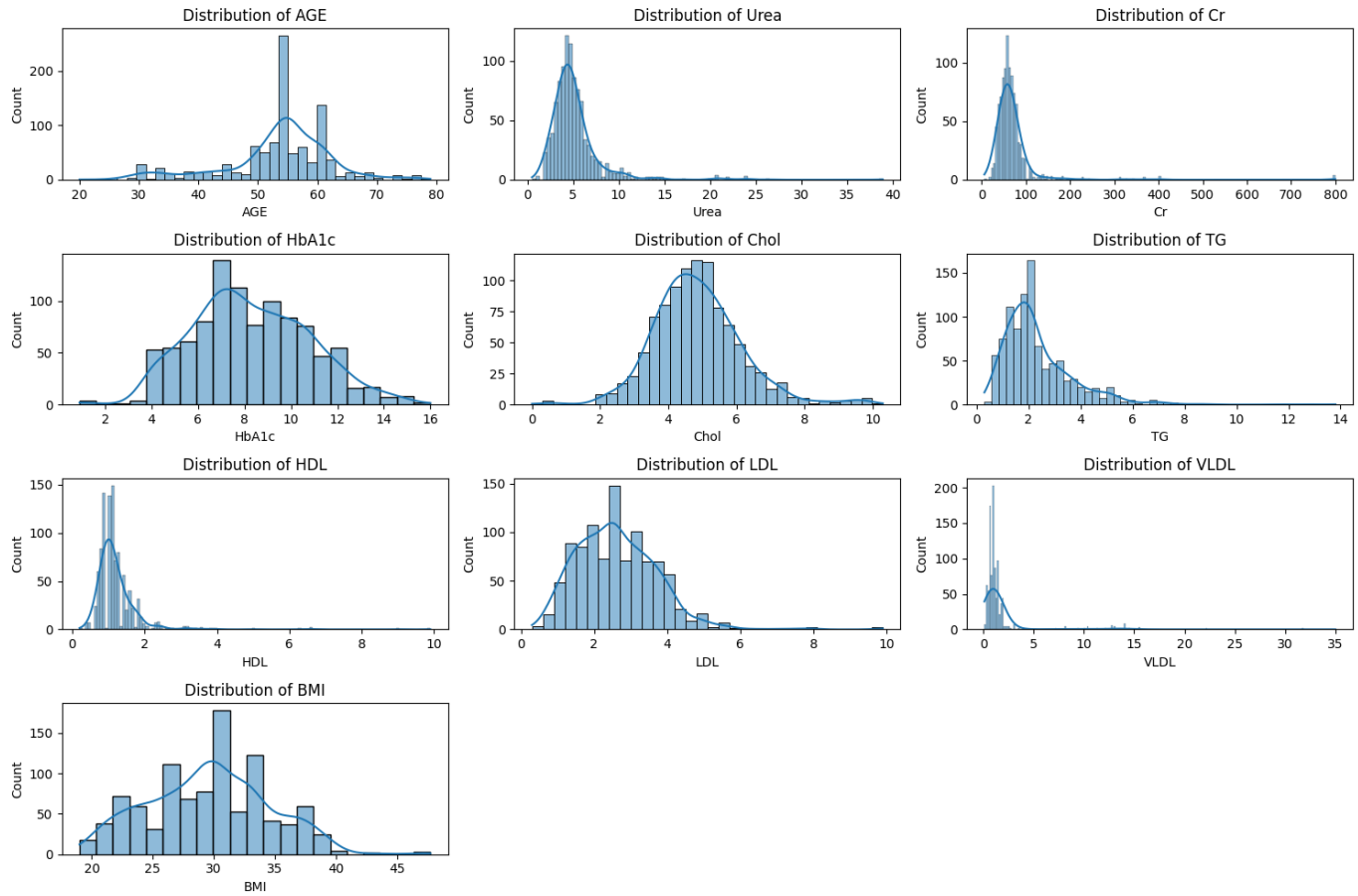

Fig 1. Distribution of Numerical Clinical Features

Histogram analysis of numerical features shows that several biochemical markers, including Urea, Creatinine (Cr), Triglycerides (TG) and VLDL exhibit right-skewed distributions. This indicates the presence of patients with elevated values, which is common in clinical datasets.

HbA1c displays a broad distribution range, which aligns with its role as a diagnostic marker for diabetes. BMI and AGE show moderate spread without extreme irregularities. Since tree-based models are robust to skewed data, no aggressive transformation was required.

5.3 Outlier Analysis

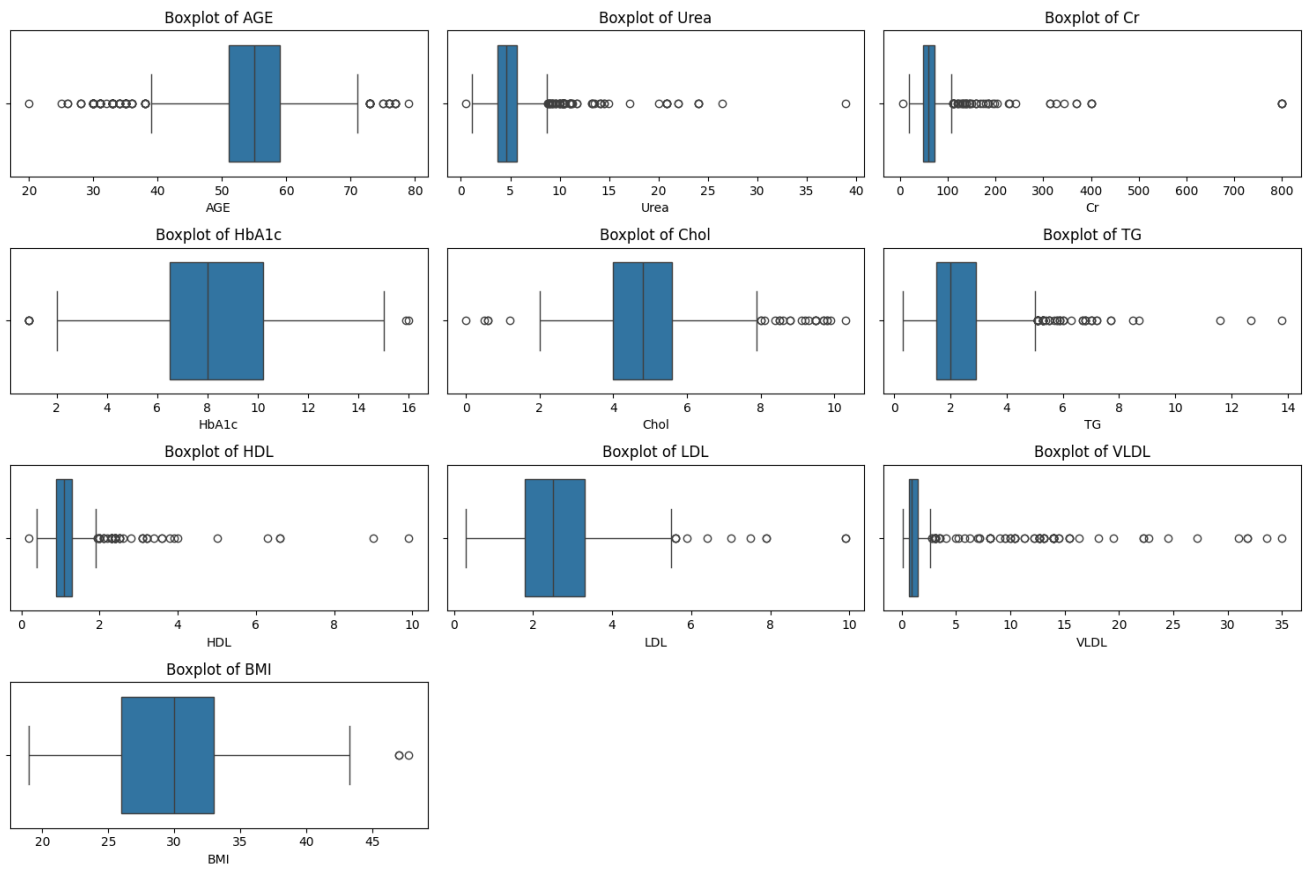

Fig 2. Outlier Analysis Using Boxplots

Boxplot analysis indicates the presence of outliers in features such as Creatinine, Urea, TG, HDL and VLDL. These extreme values likely represent clinically severe cases rather than data errors.

Outliers were retained to preserve real-world variability. Removing them could reduce the model's ability to generalize to patients with abnormal clinical measurements. Given the use of Random Forest, which is relatively insensitive to outliers, no additional outlier treatment was applied.

5.4 Correlation Analysis

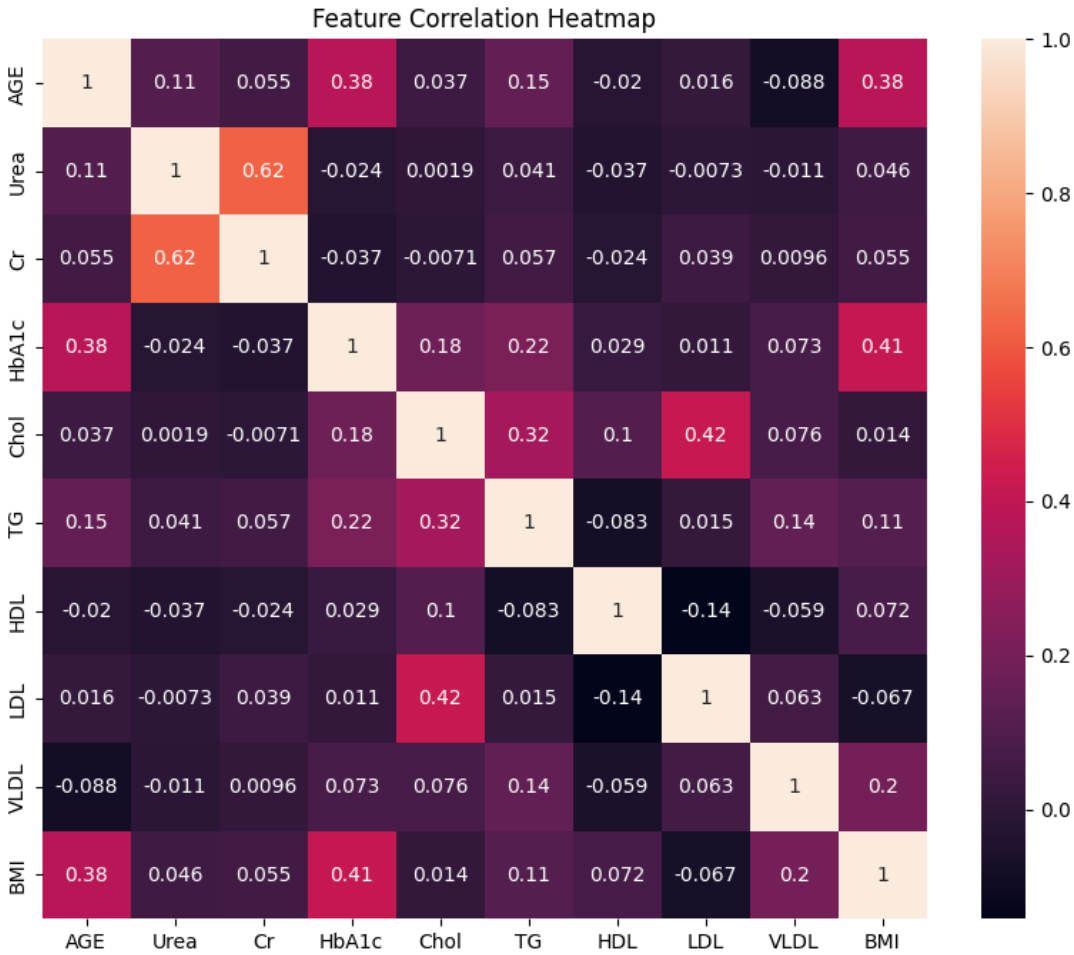

Fig 3. Correlation Heatmap of Clinical Features

The correlation heatmap shows moderate relationships among certain features. Notably, Urea and Creatinine exhibit a strong positive correlation (~0.62), reflecting their clinical association with kidney function. Cholesterol and LDL also demonstrate expected correlation (~0.42). HbA1c shows moderate correlation with BMI, suggesting metabolic linkage.

No extremely high correlations were observed, indicating the absence of severe multicollinearity. This supports stable model training without the need for feature removal.

6. Preprocessing Strategy

A *ColumnTransformer* was implemented to ensure structured and reproducible preprocessing. Numerical features were standardized using *StandardScaler*, while the categorical feature (Gender) was encoded using *OneHotEncoder*.

All preprocessing steps were integrated within a unified *Pipeline*. This approach prevents data leakage, maintains reproducibility across experiments, and ensures that preprocessing and model logic remain consistent during deployment and serialization.

7. Supervised Learning (Classification)

Two supervised models were developed and evaluated: *Logistic Regression* and *Random Forest*. Model performance was assessed using Accuracy, Precision, Recall, F1-score, and Confusion Matrix analysis.

7.1 Logistic Regression

Logistic Regression was trained using balanced class weight to address the class imbalance present in the dataset.

On the test set, the model achieved an accuracy of 93.5%, with a macro F1-score of 0.80 and a weighted F1-score of 0.94. While overall performance appears strong, class-wise evaluation reveals differences across categories.

The confusion matrix indicated that the model performs very well in identifying the majority Diabetic (Y) class and reasonably well for the Non-diabetic (N) class. However, performance on the Predict-diabetic (P) class is comparatively weaker. Several predict-diabetic instances are misclassified as either normal or diabetic, suggesting difficulty in capturing the transitional nature of this group.

This behavior is consistent with Logistic Regression's linear decision boundary, which may not fully capture complex, non-linear relationships between clinical features and diabetic categories.

7.2 Random Forest

A Random Forest classifier was trained using 200 estimators with balanced class weight.

On the test set, the model achieved an accuracy of 99.5%, with a macro F1-score of 0.98 and a weighted F1-score of 0.99, indicating highly consistent performance across all classes.

The confusion matrix showed near-perfect classification for both the Non-diabetic and Diabetic classes. The Predict-diabetic class is also identified accurately in most cases, with only minimal misclassification. Compared to Logistic Regression, there is a clear improvement in minority class detection.

This improvement suggests that Random Forest effectively captures non-linear relationships within the dataset, particularly those driven by HbA1c and other metabolic indicators.

7.3 Cross-Validation

To evaluate model robustness, 5-fold Stratified Cross-Validation was performed using weighted F1-score.

- Logistic Regression: Mean CV F1 ≈ 0.918
- Random Forest: Mean CV F1 ≈ 0.977

The consistency between cross-validation and test performance suggests that the models are stable and not overfitting to a specific train-test split.

7.4 Hyperparameter Tuning

GridSearchCV was applied to optimize model parameters.

For Logistic Regression, the regularization parameter C was tuned. The best value was C = 10, resulting in a slight performance improvement.

For Random Forest, n_estimators and max_depth were tuned. The optimal configuration was:

- n_estimators = 100
- max_depth = None

Performance remained similar to the default model, indicating that the initial configuration was already close to optimal.

7.5 Feature Importance Analysis

Feature importance derived from the Random Forest model highlights HbA1c as the most influential predictor (~42%), followed by BMI and AGE.

The dominance of HbA1c aligns with clinical understanding, as it is a primary diagnostic indicator of diabetes. This suggests that the model is learning medically meaningful patterns rather than relying on spurious correlations.

Overall, feature importance analysis supports both model validity and interpretability.

8. Unsupervised Learning (Clustering)

To examine the natural structure of the dataset independent of labels, KMeans clustering was applied on the scaled feature set. The target variable was excluded during clustering to ensure purely unsupervised learning.

8.1 Determining the Optimal Number of Clusters

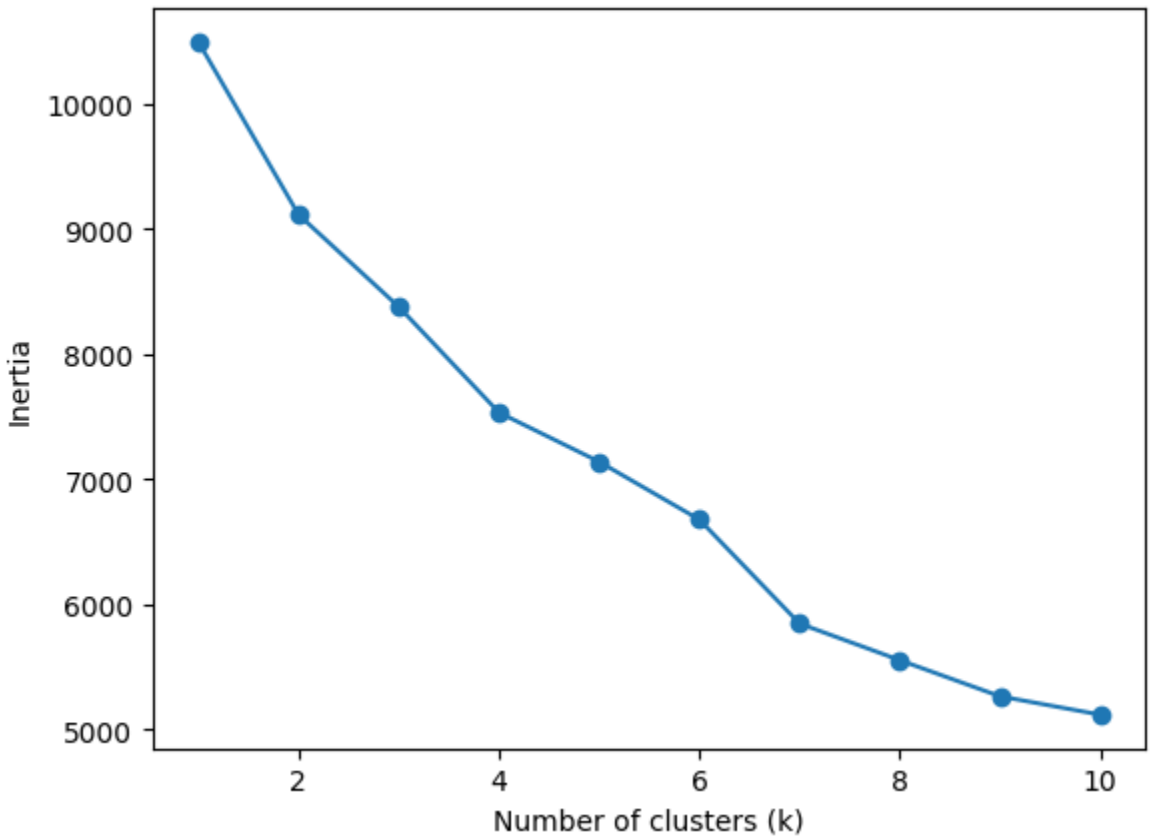

Fig 4. Elbow Method for KMeans Clustering

The Elbow plot shows a steady reduction in inertia as the number of clusters increases. A visible bend appears around k = 3-4, indicating diminishing returns beyond this range.

To support this observation, Silhouette scores were computed for k values between 2 and 6. The highest Silhouette score (0.169) was obtained at k = 2, suggesting that two clusters provide the most compact and well-separated grouping in the dataset.

Based on both inertia and Silhouette analysis, k = 2 appears to represent the most natural clustering structure.

8.2 Cluster Visualization and Interpretation

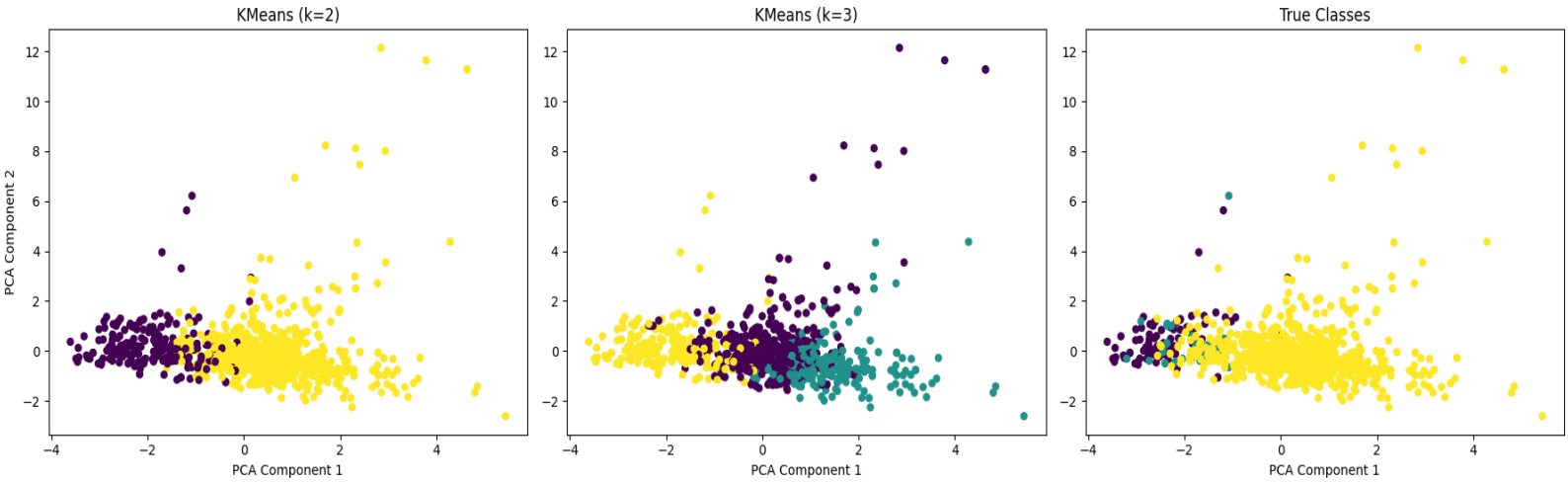

Fig 5. PCA Visualization of KMeans Clusters and True Classes

The PCA visualization compares clustering results for k = 2 and k = 3 with the true class distribution.

For k = 2, one cluster predominantly contains Diabetic (Y) cases, while the other cluster contains most Non-diabetic (N) and Predict-diabetic (P) samples along with some overlap. This indicates that the dataset naturally separates into a strong diabetic group and a mixed group.

For k = 3, the algorithm splits the data into three clusters; however, instead of isolating the Predict-diabetic class, it divides the Diabetic population into subgroups. The Predict-diabetic samples remain distributed across clusters rather than forming a distinct group.

This suggests that the dataset naturally forms two dominant clusters rather than three clearly separable clinical categories. While supervised classification can distinguish all three classes using label information, unsupervised clustering reflects the underlying overlap between Normal and Predict-diabetic cases.

9. Classification vs Clustering Comparison

Supervised classification and unsupervised clustering provide different perspectives on the same dataset.

Classification uses labeled data to learn explicit decision boundaries between classes. In this project, supervised models were able to distinguish among all three categories (Non-diabetic, Prediabetic, and Diabetic) with high accuracy, particularly using Random Forest.

Clustering, on the other hand, does not use label information. It identifies natural groupings based solely on feature similarity. The clustering results showed that the data naturally forms two dominant groups rather than three clearly separable clusters. The Diabetic class formed a strong and distinct cluster, while Non-diabetic and Predict-diabetic samples overlapped significantly.

This comparison highlights an important insight: although three clinical categories exist, the underlying feature space suggests that Predict-diabetic behaves as a transitional state between Normal and Diabetic conditions. Supervised learning is therefore necessary to learn finer boundaries that clustering alone cannot capture.

10. Error Analysis

The primary classification errors were observed in the Predict-diabetic (P) class. Logistic Regression struggled to accurately detect this minority group, often misclassifying these samples as either Normal or Diabetic. This reflects both the overlapping nature of the feature space and the limitations of a linear decision boundary. Random Forest reduced these errors significantly, indicating its ability to capture non-linear relationships more effectively.

Key observations:

- Most misclassifications occurred between Predict-diabetic (P) and Normal (N) classes.
- Logistic Regression showed lower recall for the Predict-diabetic class.
- Random Forest improved minority class detection substantially.
- Class imbalance likely contributed to residual bias toward the majority class.
- Error patterns align with clustering results, where Predict-diabetic cases did not form a distinct group.

Overall, the misclassification trends reinforce the interpretation that Prediabetes represents a transitional state between Normal and Diabetic categories.

11. Limitations

While the models achieved strong performance, several limitations exist within the dataset and experimental setup. These factors may influence generalizability and model robustness.

The key limitations are listed below:

- Severe class imbalance, with the Diabetic class dominating the dataset.
- Very limited number of Predict-diabetic samples, reducing statistical representation.
- Overlapping feature distributions between Normal and Predict-diabetic cases.
- Clustering quality constrained by lack of clearly separable natural groups.
- Absence of an external validation dataset to evaluate generalization.

These constraints should be considered when interpreting the results and assessing real-world applicability.

12. Future Enhancements

Several improvements could further strengthen the robustness, interpretability and fairness of the model.

Potential enhancements are listed below:

- Applying advanced imbalance handling techniques such as SMOTE.
- Comparing additional ensemble models such as Gradient Boosting or XGBoost.
- Incorporating SHAP-based explainability for feature-level interpretation.
- Performing probability calibration for improved confidence estimation.
- Evaluating performance using One-vs-Rest ROC-AUC.
- Expanding the dataset, particularly for the Prediabetic class, to improve balance.

These enhancements would improve both predictive stability and clinical interpretability.

13. Conclusion

This project demonstrates a complete and structured machine learning workflow, beginning with data cleaning and exploratory analysis, followed by robust preprocessing, model development, cross-validation, hyperparameter tuning, and detailed interpretation of results.

The supervised learning models showed strong predictive performance, with Random Forest achieving highly robust and consistent results across all classes. Its ability to capture non-linear feature interactions significantly improved detection of minority classes, particularly the Predict-diabetic group. Feature importance analysis further validated that the model relied on clinically

meaningful indicators such as HbA1c, BMI, and AGE, reinforcing alignment between data-driven findings and medical knowledge.

In contrast, unsupervised clustering provided insight into the intrinsic structure of the dataset. The clustering results indicated that the data naturally forms two dominant groups rather than three clearly separable categories. This suggests that Predict-diabetic behaves as a transitional metabolic state rather than a sharply defined cluster, which explains the classification challenges observed in error analysis.

The comparison between supervised and unsupervised approaches highlights an important takeaway: while natural groupings reveal underlying data structure, supervised learning is essential for learning precise diagnostic boundaries when class labels are available.

Overall, the project not only achieves high predictive accuracy but also demonstrates analytical reasoning, careful validation, and meaningful interpretation of results. The findings provide both strong modeling performance and valuable insight into the clinical characteristics of diabetic progression.